R84/B_attention_direct | Locked full18 results

Same unique VAL checkpoint on VAL/TEST/LIB; native/FP64 ind; historical NSS weights disclosed.

epoch 26 | update 37440 | seed2 | fresh initialization

Data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
VAL ALL	71.0000	66.2037	11.416697	11.427978	11.371692	-0.1927	+0.3949	0.4050
TEST ALL	71.2235	66.1802	11.373993	11.385299	11.329932	-0.1975	+0.3857	0.3946

Dataset	n	NSS Top1 (%)	R84 Top1 (%)	NSS cost	R84 cost
TSP TEST	4500	45.888889	46.266667	7.970400719	7.970534862
CVRP TEST	4500	45.977778	47.155556	18.004802368	18.016121155
TSPLIB	49	20.408163	28.571429	8.217907074	8.205368125
CVRPLIB	100	70.000000	61.000000	67.101111717	67.284313650

NSS: R79 weights trained on historical70114 TRAIN, selected using historical VAL; not retrained on new90k.

NSS TEST predictions were freshly computed on these4500 instances; old3895 TEST metrics are not reused.

SBS is fitted on the new81000 TRAIN only. Oracle is the per-instance legal-pool minimum.

ALL is the equal-problem mean; mixed raw cost is descriptive. TEST informed prior research.

Method selection and true winner use separate colors on the same axes; Top1 is per-instance accuracy.

Selection head: direct_attention | focal extra: False

R84/B_attention_direct | VAL: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	71.0000	66.2037	11.416697	11.427978	11.371692	-0.1927	+0.3949	0.4050	per-problem TRAIN SBS
TSP	46.0889	33.1111	8.025089	8.045354	7.990533	-0.2519	+0.4325	0.4209	T2T
CVRP	46.8000	18.7111	18.160705	18.269377	18.064087	-0.5948	+0.5349	0.7136	RELD_CVRP
ATSP	76.2889	47.4889	1.638811	1.676905	1.632471	-2.2717	+0.3884	0.3852	UNICO_MatPOENet
OVRP	41.2000	35.7556	7.461891	7.469579	7.415777	-0.1029	+0.6218	0.6331	MoSES_RF
VRPB	81.2000	80.8444	9.355699	9.354876	9.318750	+0.0088	+0.3965	0.4002	reld_mt
VRPL	36.6889	33.0889	12.431892	12.437959	12.383444	-0.0488	+0.3912	0.3976	MoSES_RF
VRPTW	80.0222	79.9556	18.818896	18.819431	18.766139	-0.0028	+0.2811	0.2972	reld_mt
OVRPTW	83.8444	83.7778	10.143252	10.143730	10.116859	-0.0047	+0.2609	0.2580	reld_mt
OVRPB	88.4889	88.5111	6.357212	6.357254	6.341078	-0.0007	+0.2544	0.2639	reld_mt
OVRPL	41.9778	35.9111	7.437279	7.445709	7.390416	-0.1132	+0.6341	0.6498	MoSES_RF
VRPBL	78.7778	78.5111	9.369513	9.371523	9.328366	-0.0214	+0.4411	0.4451	reld_mt
VRPBTW	78.6000	78.4222	19.945495	19.950430	19.851084	-0.0247	+0.4756	0.4554	reld_mt
VRPLTW	78.8889	78.7556	18.763796	18.765871	18.707843	-0.0111	+0.2991	0.3140	reld_mt
OVRPBL	89.3333	89.2000	6.368994	6.370093	6.352824	-0.0173	+0.2545	0.2650	reld_mt
OVRPBTW	83.6000	83.4444	10.661777	10.661098	10.623245	+0.0064	+0.3627	0.3405	reld_mt
OVRPLTW	83.5111	83.4889	10.157317	10.157423	10.129818	-0.0010	+0.2715	0.2746	reld_mt
VRPBLTW	79.5556	79.4000	19.792184	19.797044	19.706420	-0.0246	+0.4352	0.4236	reld_mt
OVRPBLTW	83.1333	83.2889	10.610752	10.609955	10.571302	+0.0075	+0.3732	0.3523	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

R84/B_attention_direct | TEST: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	71.2235	66.1802	11.373993	11.385299	11.329932	-0.1975	+0.3857	0.3946	per-problem TRAIN SBS
TSP	46.2667	34.5333	7.970535	7.989621	7.937593	-0.2389	+0.4150	0.4076	T2T
CVRP	47.1556	17.9333	18.016121	18.124983	17.918509	-0.6006	+0.5448	0.7102	RELD_CVRP
ATSP	76.6000	47.2444	1.635385	1.674207	1.629441	-2.3188	+0.3648	0.3639	UNICO_MatPOENet
OVRP	40.6667	33.6444	7.445324	7.455750	7.400487	-0.1398	+0.6059	0.6170	MoSES_RF
VRPB	80.3111	80.4000	9.351906	9.351810	9.315298	+0.0010	+0.3930	0.3995	reld_mt
VRPL	37.0000	32.2667	12.349511	12.356138	12.300958	-0.0536	+0.3947	0.4018	MoSES_RF
VRPTW	81.2000	80.8444	18.688631	18.690545	18.637408	-0.0102	+0.2748	0.2849	reld_mt
OVRPTW	83.9333	83.9333	10.092599	10.093063	10.068135	-0.0046	+0.2430	0.2343	reld_mt
OVRPB	88.2222	88.1333	6.369809	6.370826	6.352225	-0.0160	+0.2768	0.2850	reld_mt
OVRPL	42.4444	34.6000	7.463235	7.472542	7.418033	-0.1245	+0.6094	0.6194	MoSES_RF
VRPBL	79.4667	79.4222	9.395089	9.396379	9.355967	-0.0137	+0.4182	0.4159	reld_mt
VRPBTW	78.7333	78.7556	19.666895	19.665725	19.577666	+0.0059	+0.4558	0.4521	reld_mt
VRPLTW	79.2222	79.2444	18.801933	18.803136	18.745143	-0.0064	+0.3030	0.3233	reld_mt
OVRPBL	89.2222	89.2889	6.375832	6.375741	6.359310	+0.0014	+0.2598	0.2713	reld_mt
OVRPBTW	84.9333	84.8222	10.659098	10.659591	10.624748	-0.0046	+0.3233	0.2955	reld_mt
OVRPLTW	82.9778	82.8667	10.085486	10.086112	10.059001	-0.0062	+0.2633	0.2636	reld_mt
VRPBLTW	79.9556	79.7333	19.702676	19.707068	19.613284	-0.0223	+0.4558	0.4385	reld_mt
OVRPBLTW	83.7111	83.5778	10.661810	10.662149	10.625574	-0.0032	+0.3410	0.3188	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

R84/B_attention_direct | LIB and locked provenance

Current R84 factory, same buffers and selected checkpoint; no solver execution or LIB tuning.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
TSPLIB	28.5714	10.2041	8.205368	8.240435	8.142167	-0.4256	+0.7762	0.7666
CVRPLIB	61.0000	45.0000	67.284314	67.245364	66.945093	+0.0579	+0.5067	0.6185

SHA256: 309b90d7a1bc5470465140e3f2d9d73b0745debc2c3bf2c2907f1cf5fb79f9d1

Unique selection: strict minimum full VAL macro_vs_sbs_pct with common new81000 TRAIN-fitted SBS; first wins ties

Inference: eval mode, direct attention logits, stable ascending-global-solver-ID tie break

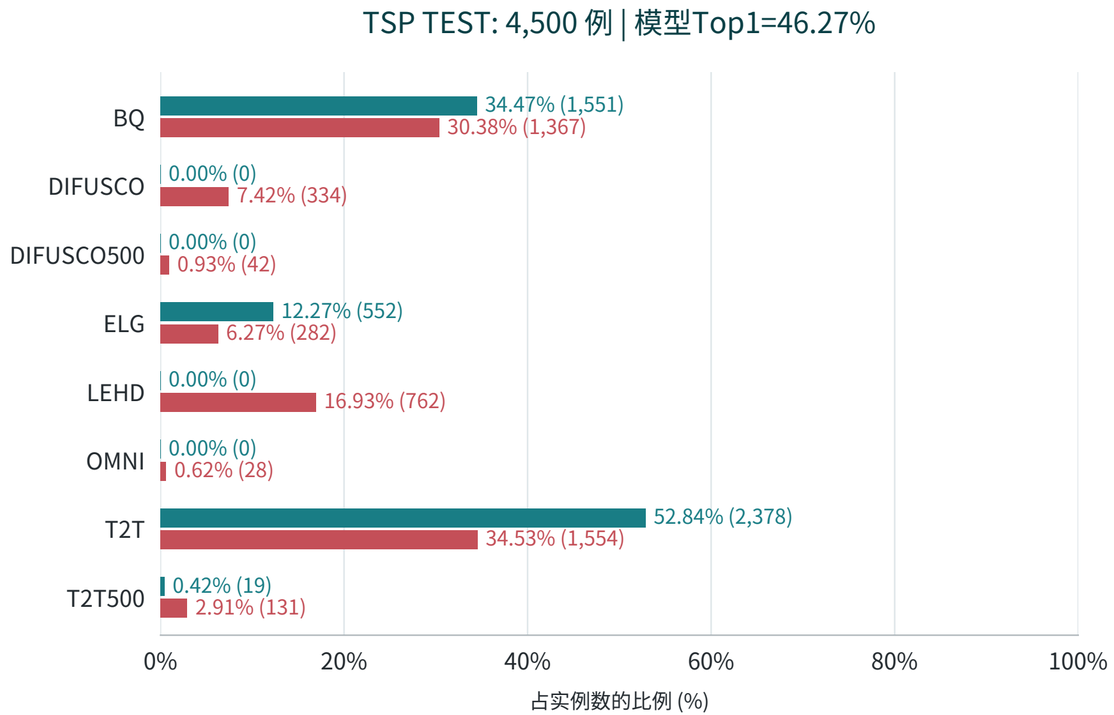
Normalization: new81000 TRAIN only; no TEST/LIB refit.

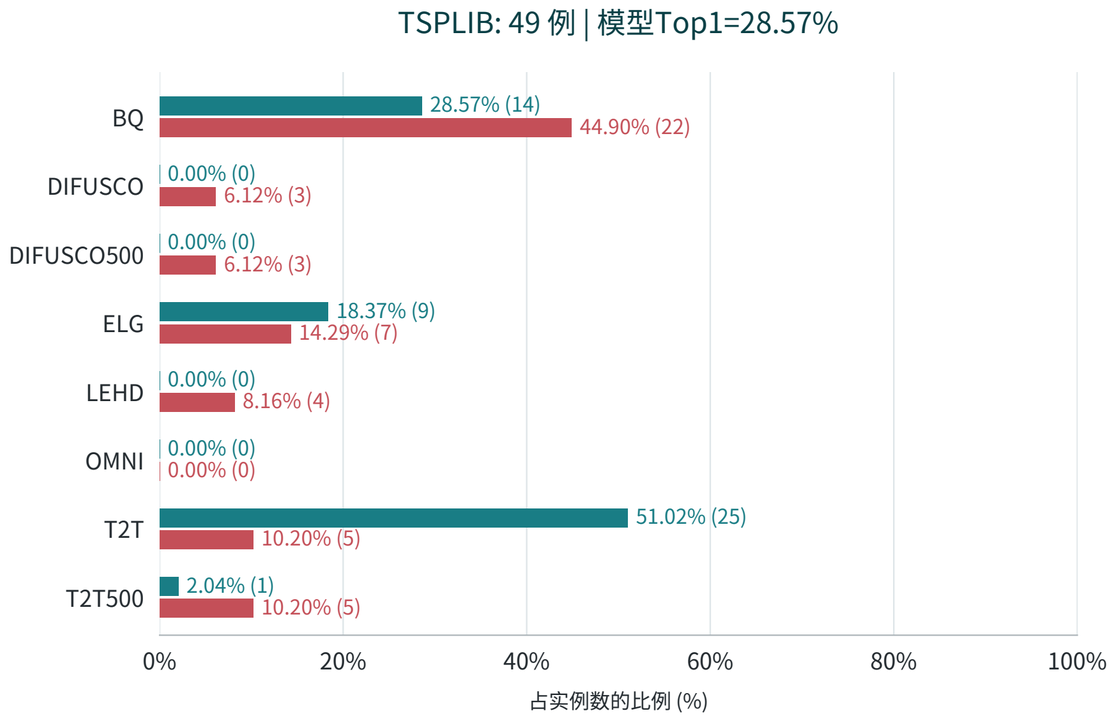
R84/B_attention_direct | TSP | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 309b90d7a1bc5470

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/TSPLIB | native/FP64 ind





标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

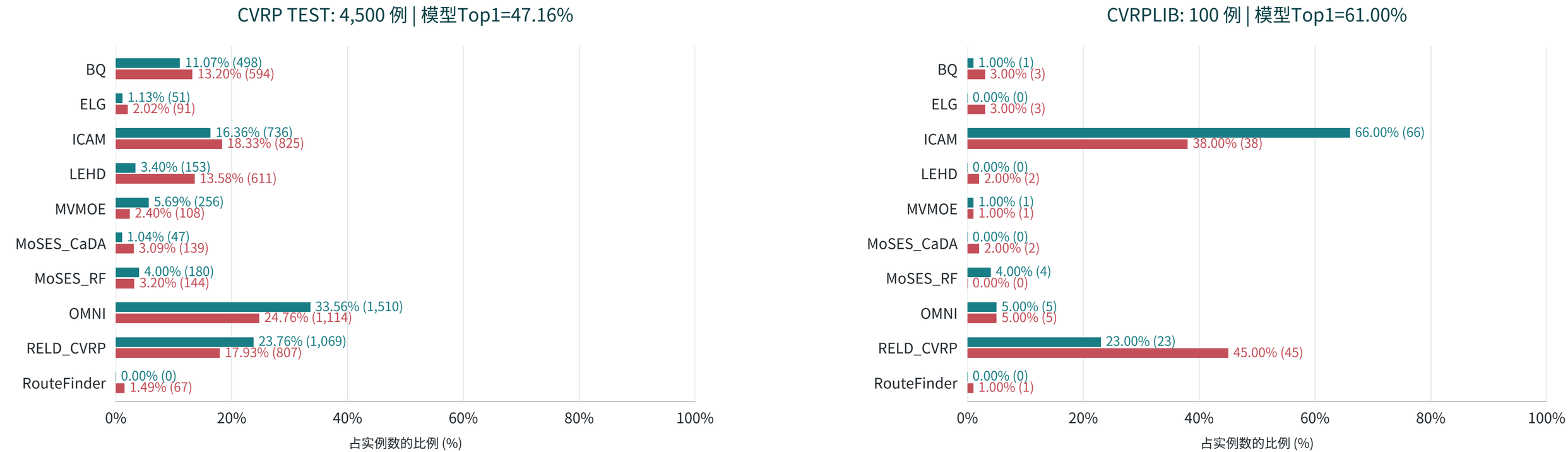
R84/B_attention_direct: stable global solver-ID score ties. Same R84 lock; LIB inference only; no training/reselection/Rxx mixing.

R84/B_attention_direct | CVRP | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 309b90d7a1bc5470

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/CVRPLIB | native/FP64 ind



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

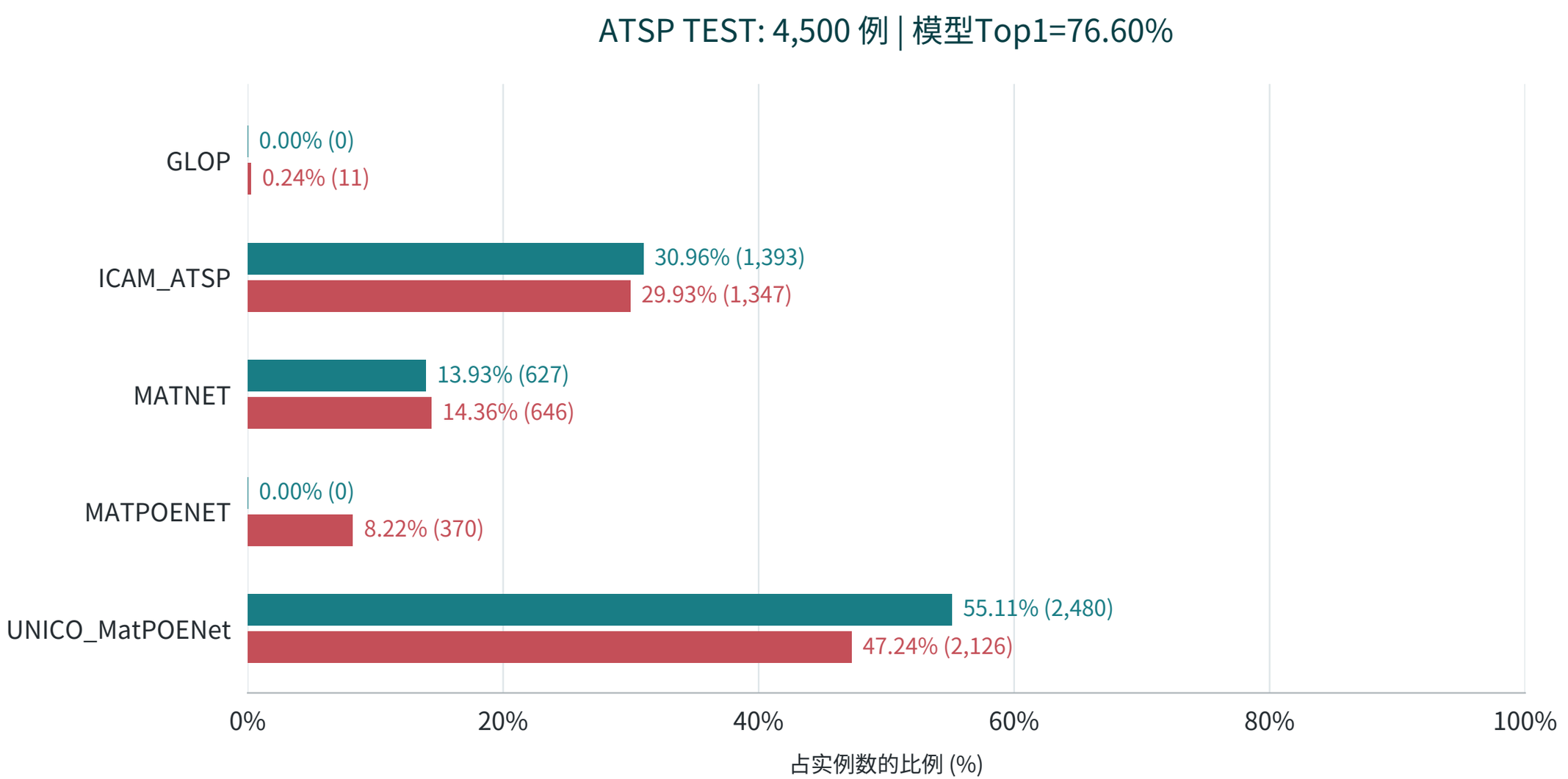
R84/B_attention_direct: stable global solver-ID score ties. Same R84 lock; LIB inference only; no training/reselection/Rxx mixing.

R84/B_attention_direct | ATSP | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 309b90d7a1bc5470

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

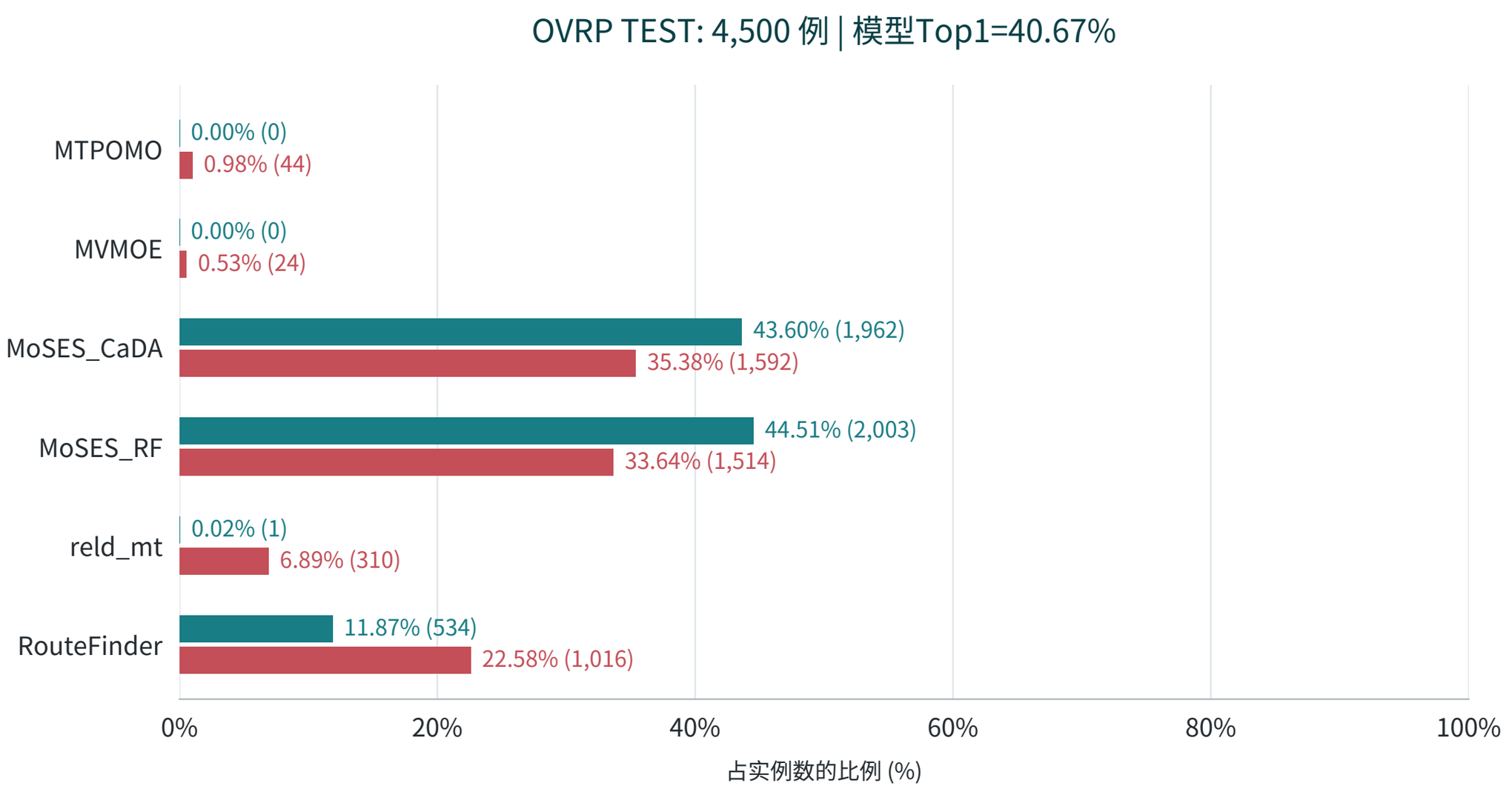
R84/B_attention_direct: stable global solver-ID score ties. Same R84 lock; LIB inference only; no training/reselection/Rxx mixing.

R84/B_attention_direct | OVRP | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 309b90d7a1bc5470

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

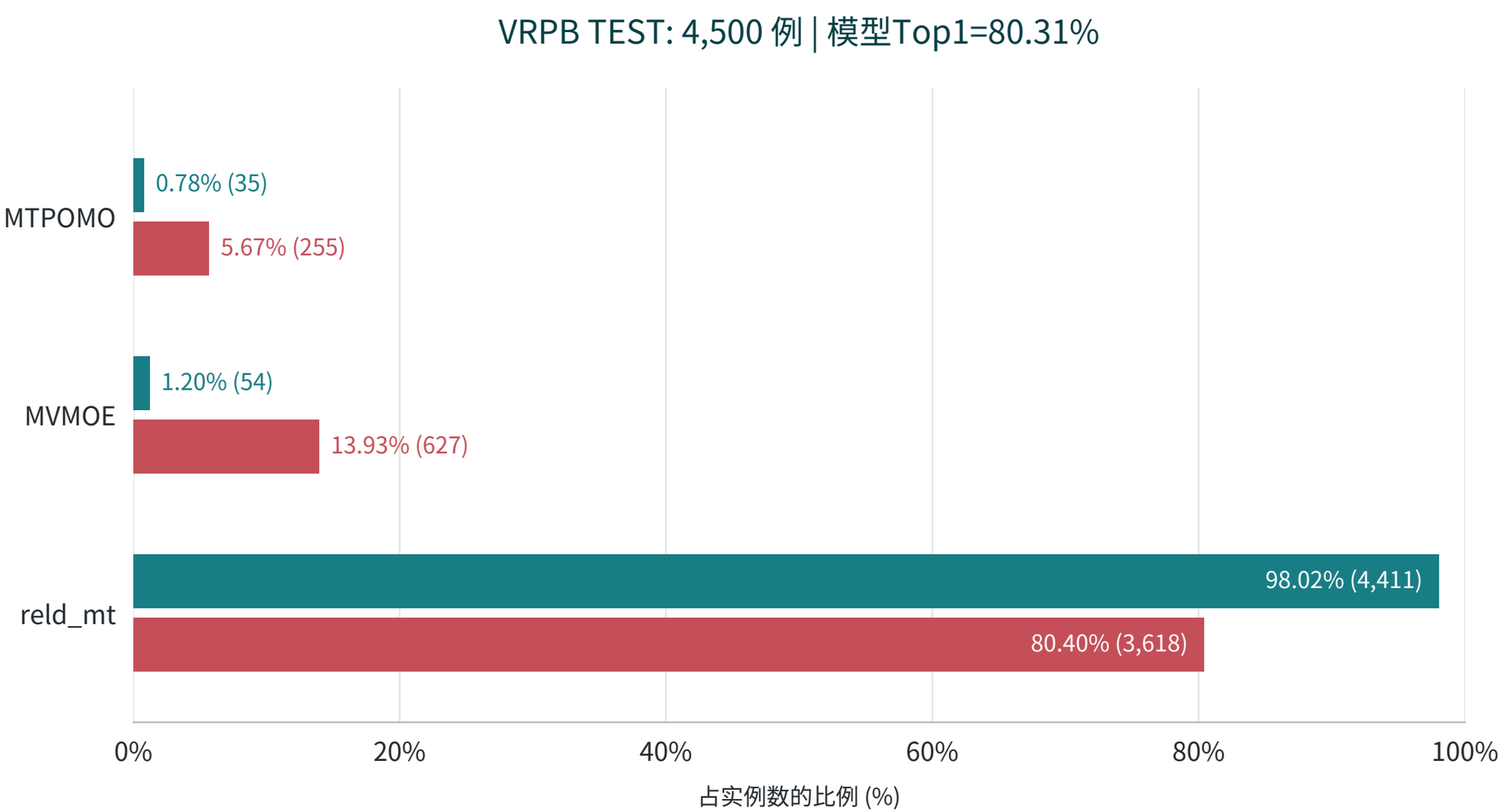
R84/B_attention_direct: stable global solver-ID score ties. Same R84 lock; LIB inference only; no training/reselection/Rxx mixing.

R84/B_attention_direct | VRPB | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 309b90d7a1bc5470

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

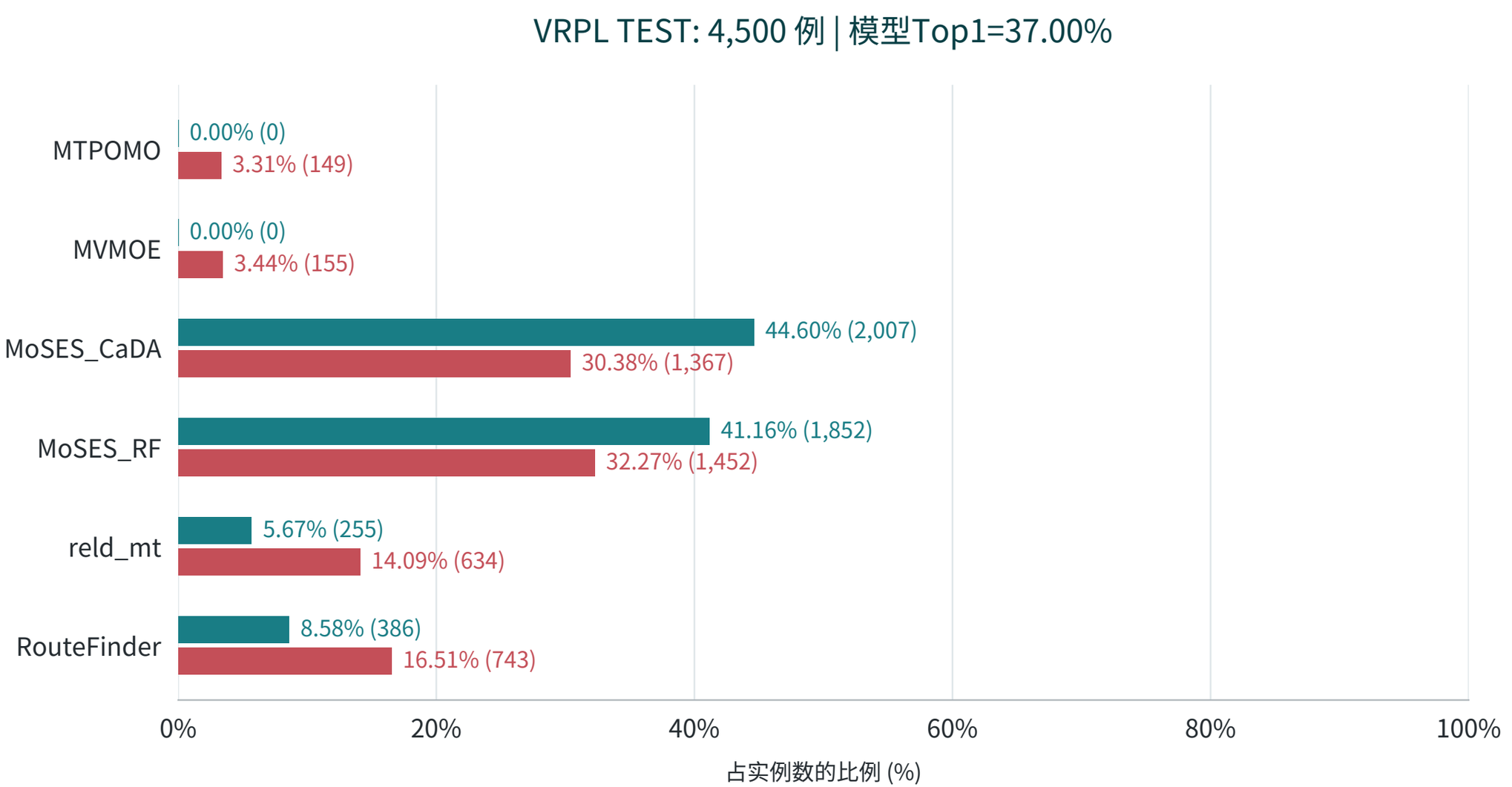
R84/B_attention_direct: stable global solver-ID score ties. Same R84 lock; LIB inference only; no training/reselection/Rxx mixing.

R84/B_attention_direct | VRPL | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 309b90d7a1bc5470

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

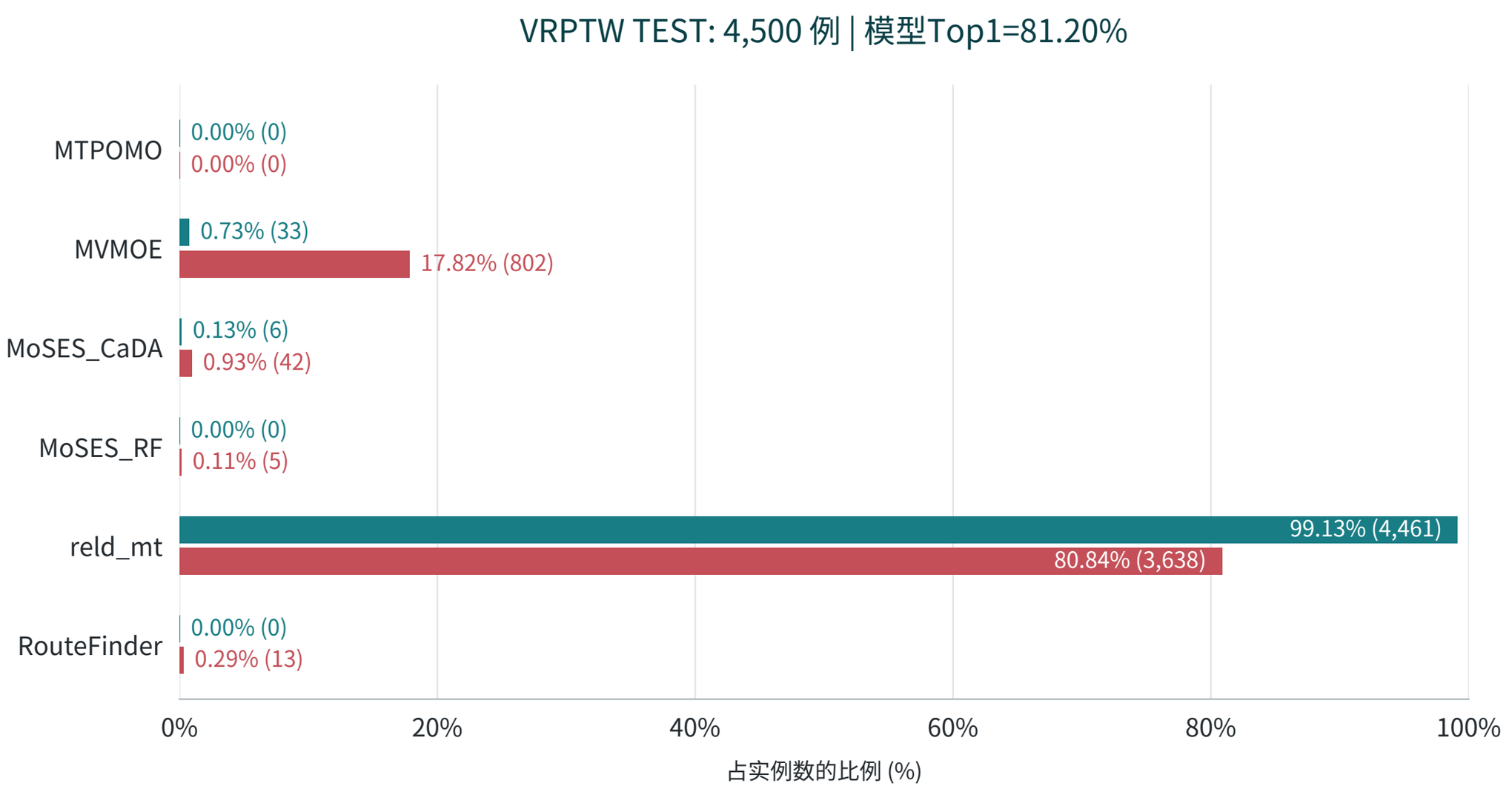
R84/B_attention_direct: stable global solver-ID score ties. Same R84 lock; LIB inference only; no training/reselection/Rxx mixing.

R84/B_attention_direct | VRPTW | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 309b90d7a1bc5470

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

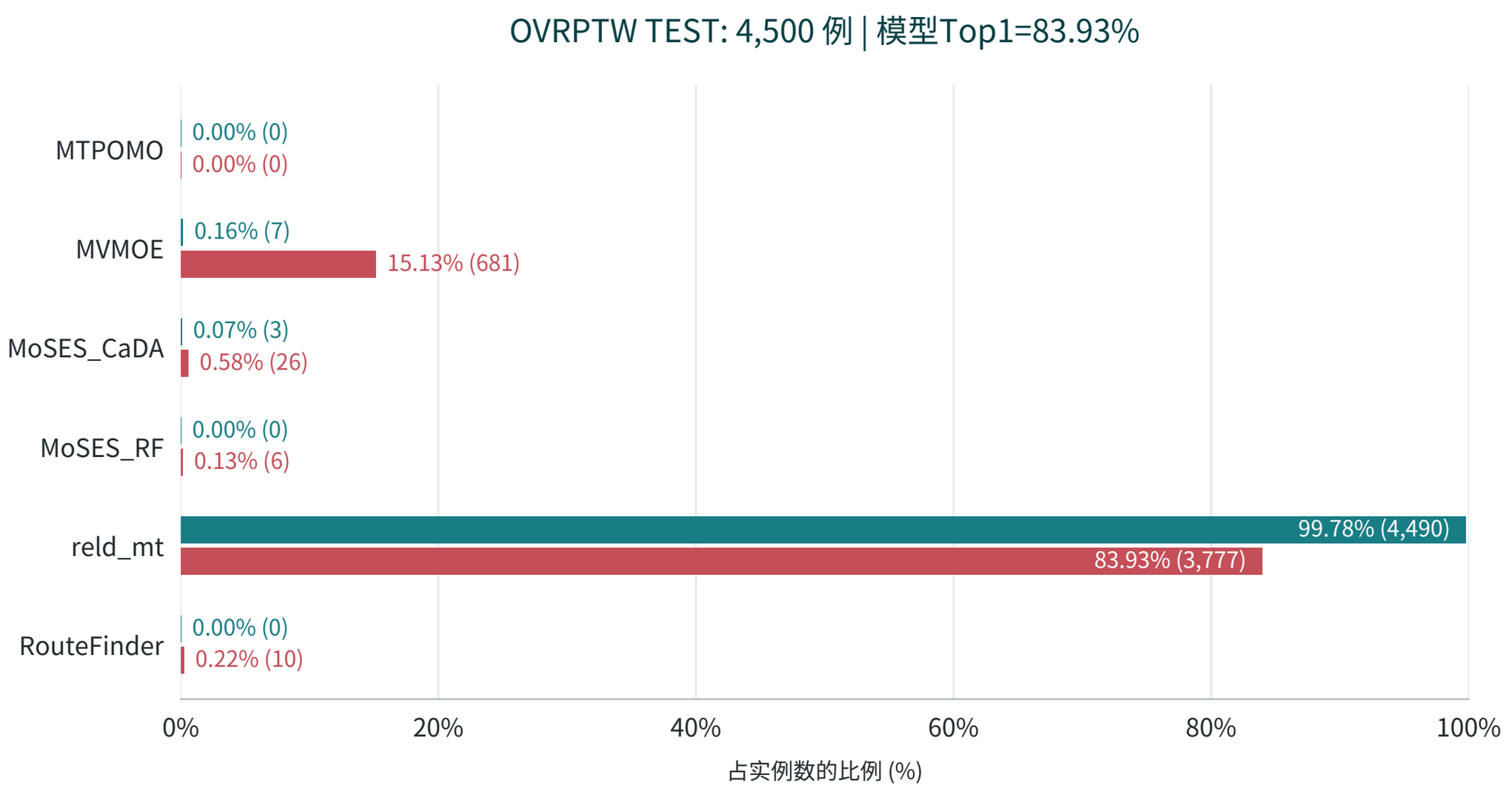
R84/B_attention_direct: stable global solver-ID score ties. Same R84 lock; LIB inference only; no training/reselection/Rxx mixing.

R84/B_attention_direct | OVRPTW | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 309b90d7a1bc5470

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

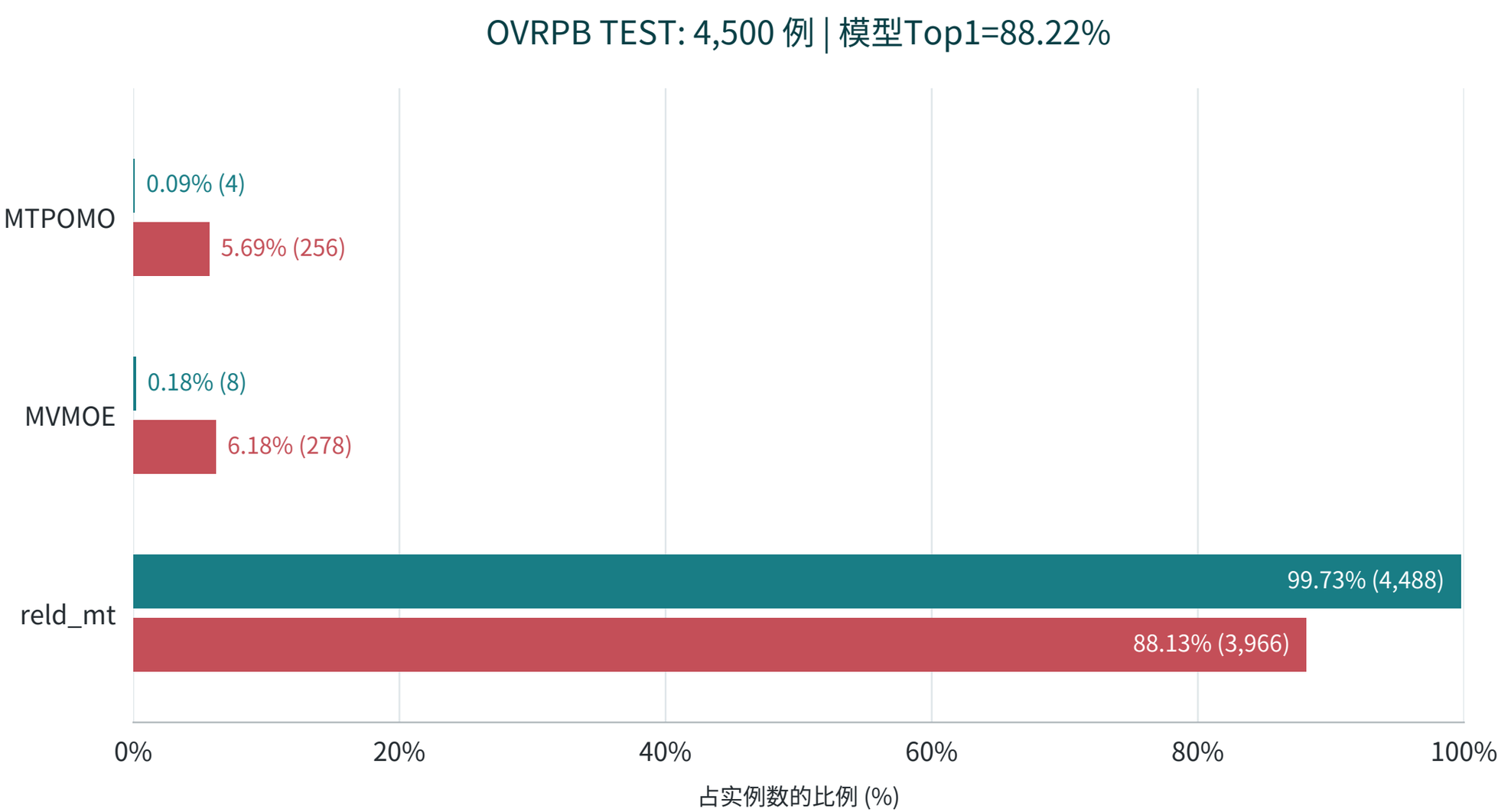
R84/B_attention_direct: stable global solver-ID score ties. Same R84 lock; LIB inference only; no training/reselection/Rxx mixing.

R84/B_attention_direct | OVRPB | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 309b90d7a1bc5470

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

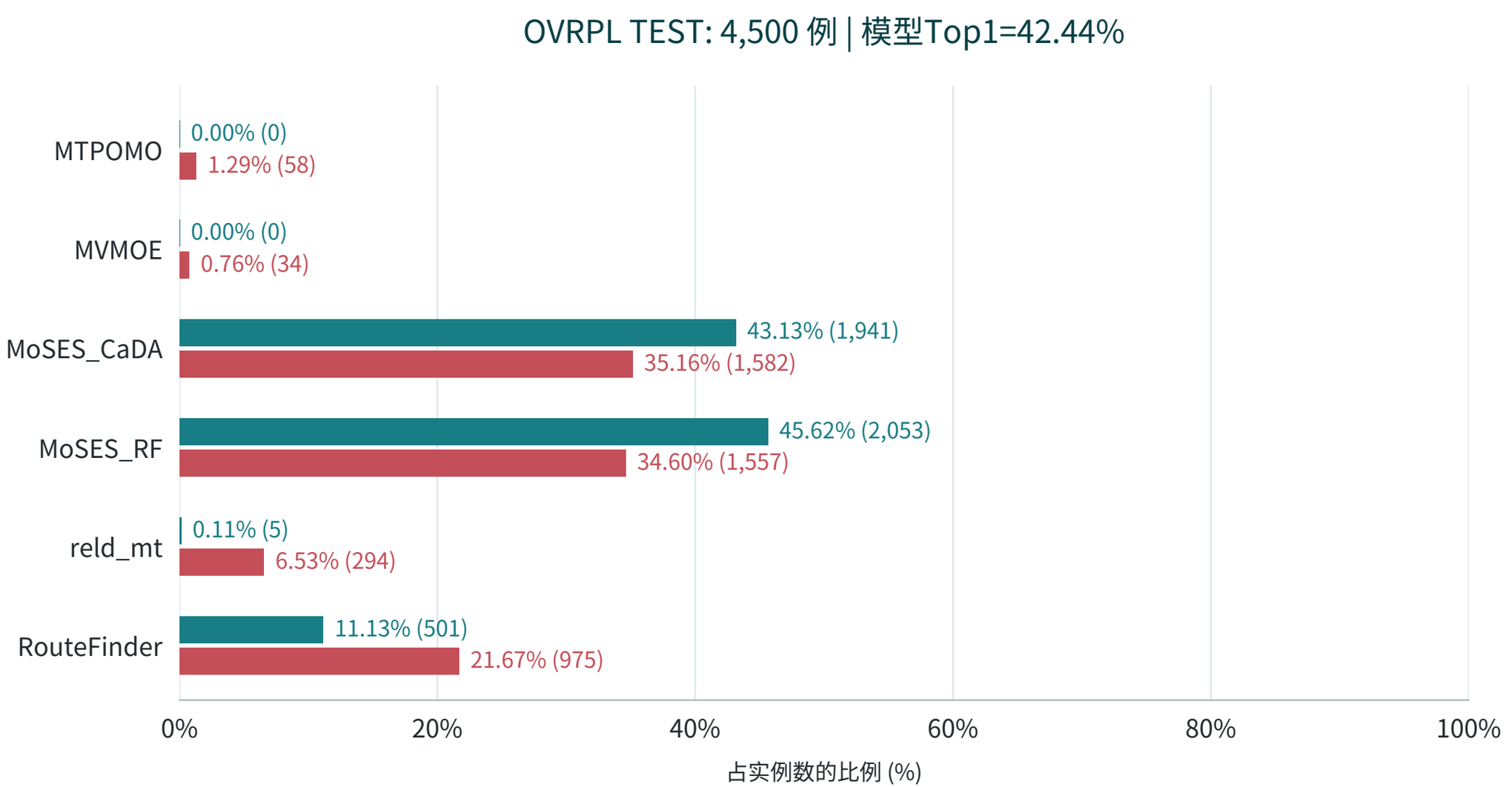
R84/B_attention_direct: stable global solver-ID score ties. Same R84 lock; LIB inference only; no training/reselection/Rxx mixing.

R84/B_attention_direct | OVRPL | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 309b90d7a1bc5470

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

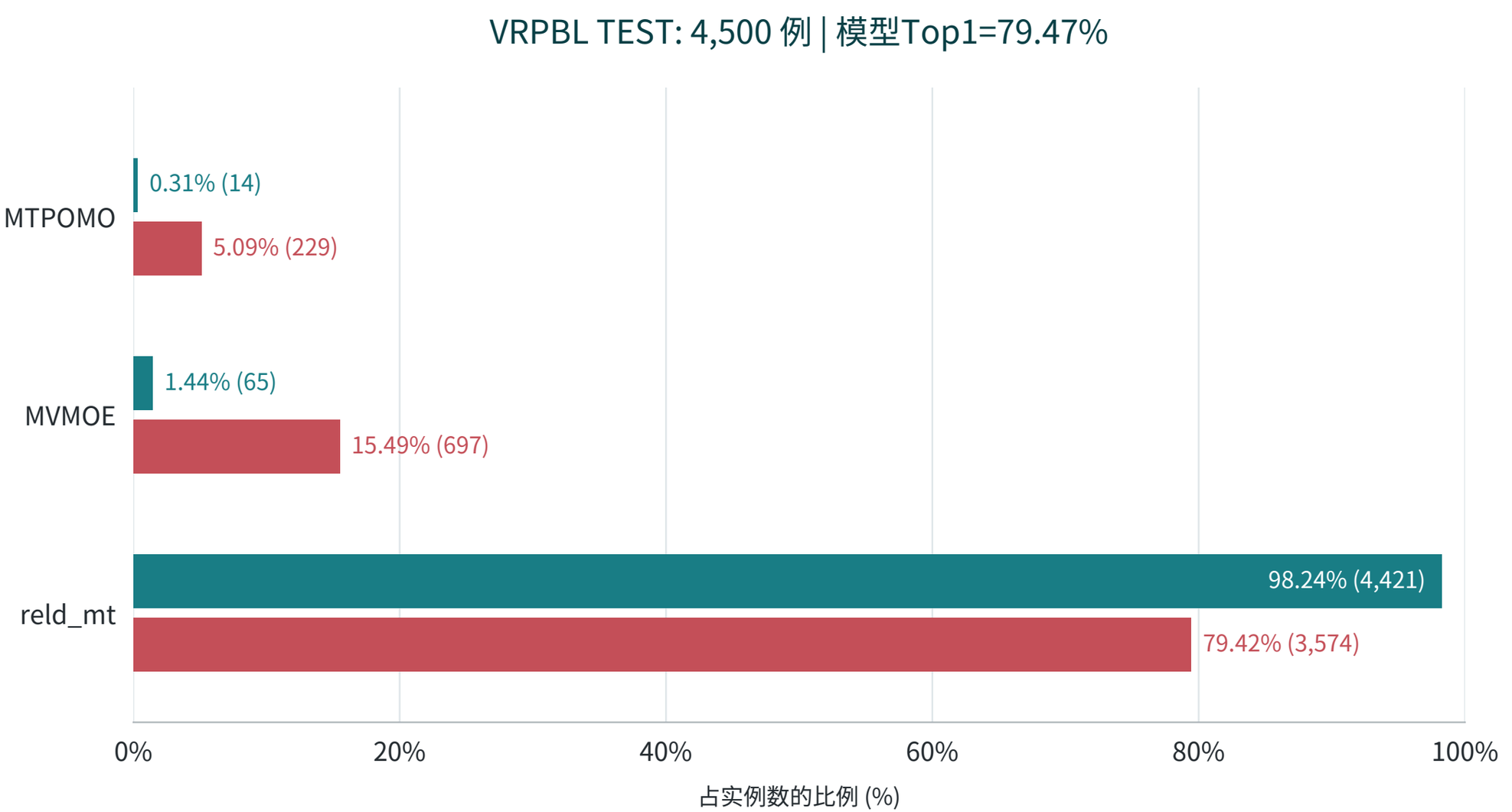
R84/B_attention_direct: stable global solver-ID score ties. Same R84 lock; LIB inference only; no training/reselection/Rxx mixing.

R84/B_attention_direct | VRPBL | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 309b90d7a1bc5470

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

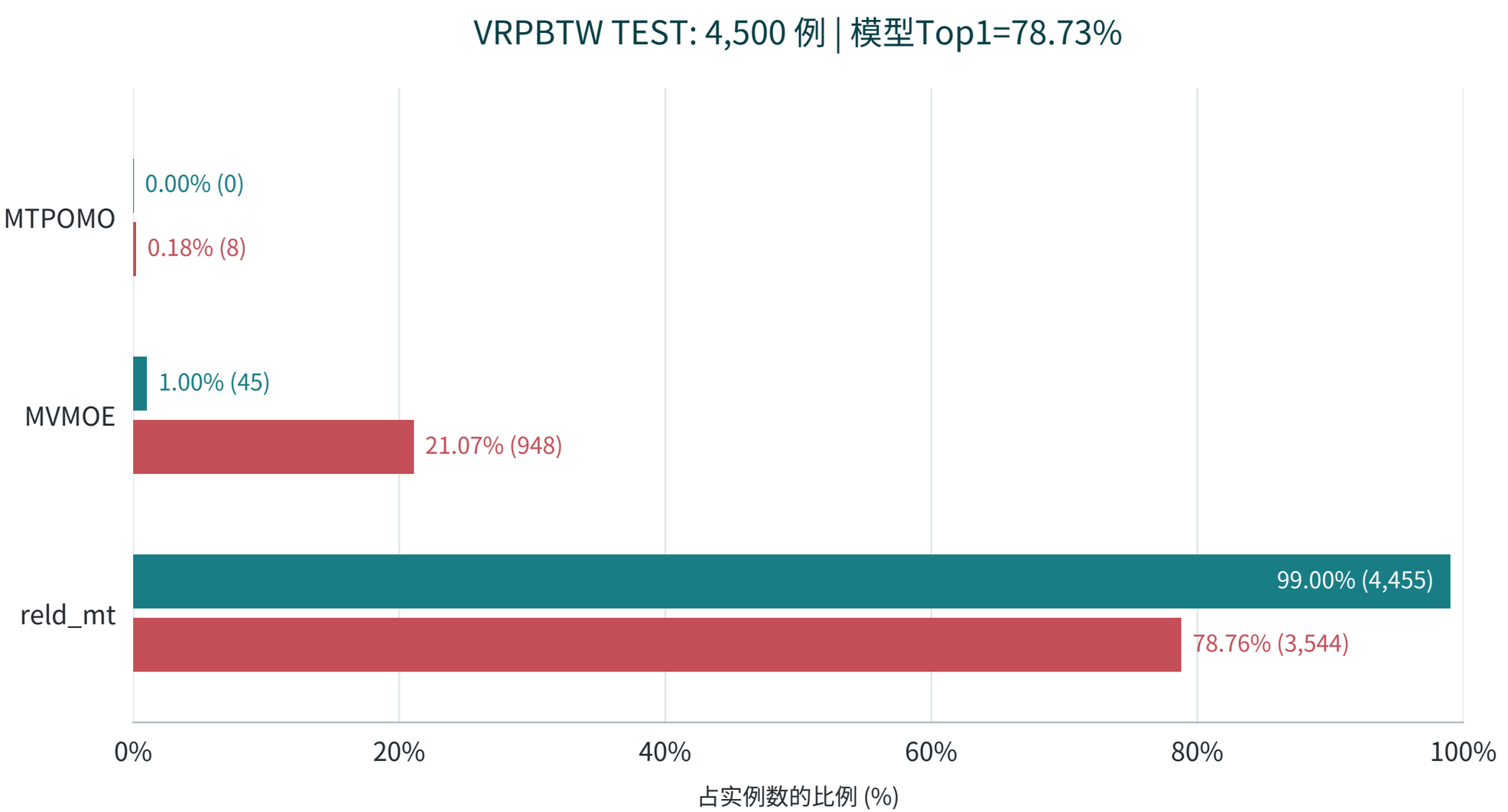
R84/B_attention_direct: stable global solver-ID score ties. Same R84 lock; LIB inference only; no training/reselection/Rxx mixing.

R84/B_attention_direct | VRPBTW | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 309b90d7a1bc5470

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

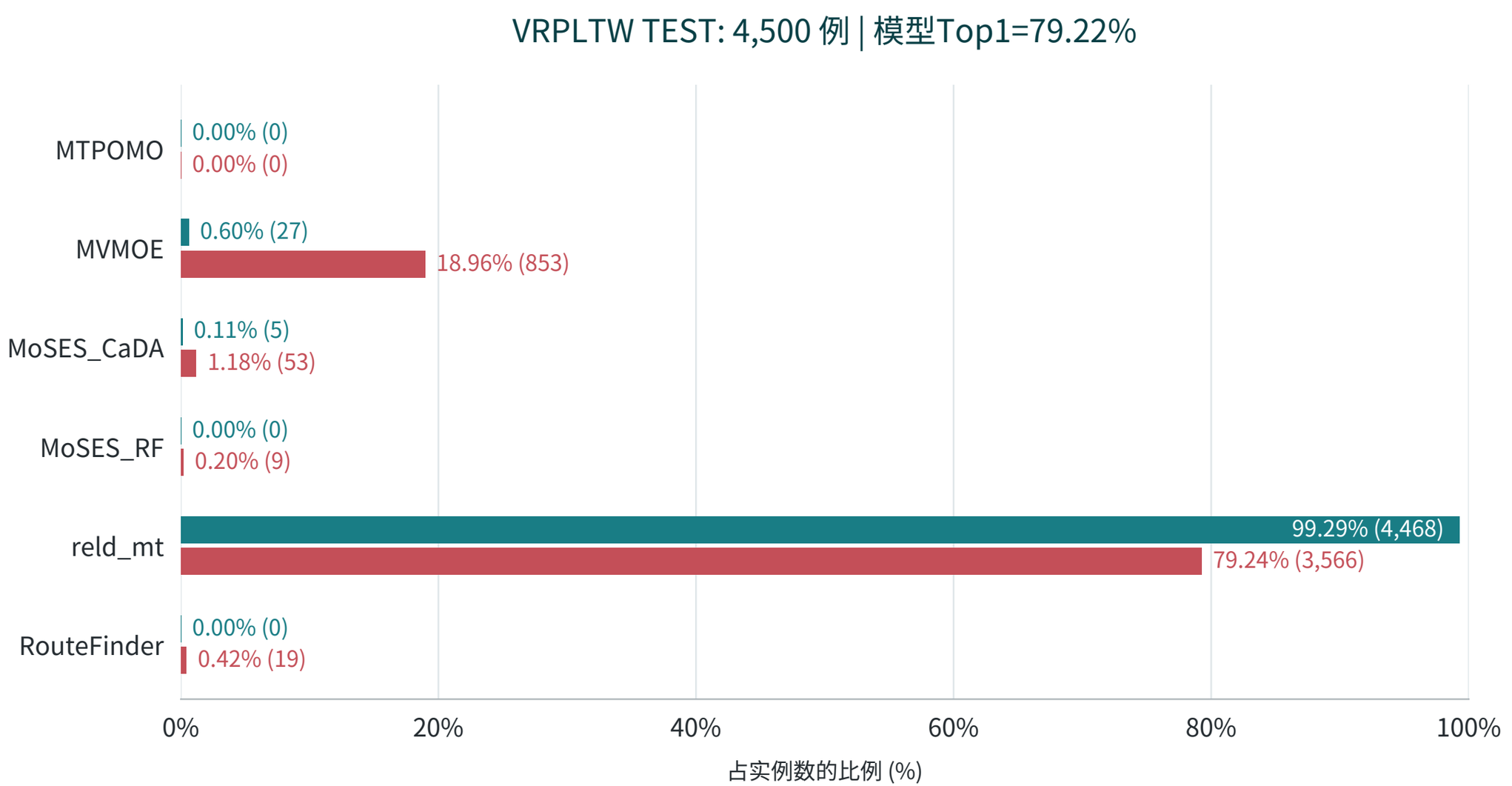
R84/B_attention_direct: stable global solver-ID score ties. Same R84 lock; LIB inference only; no training/reselection/Rxx mixing.

R84/B_attention_direct | VRPLTW | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 309b90d7a1bc5470

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

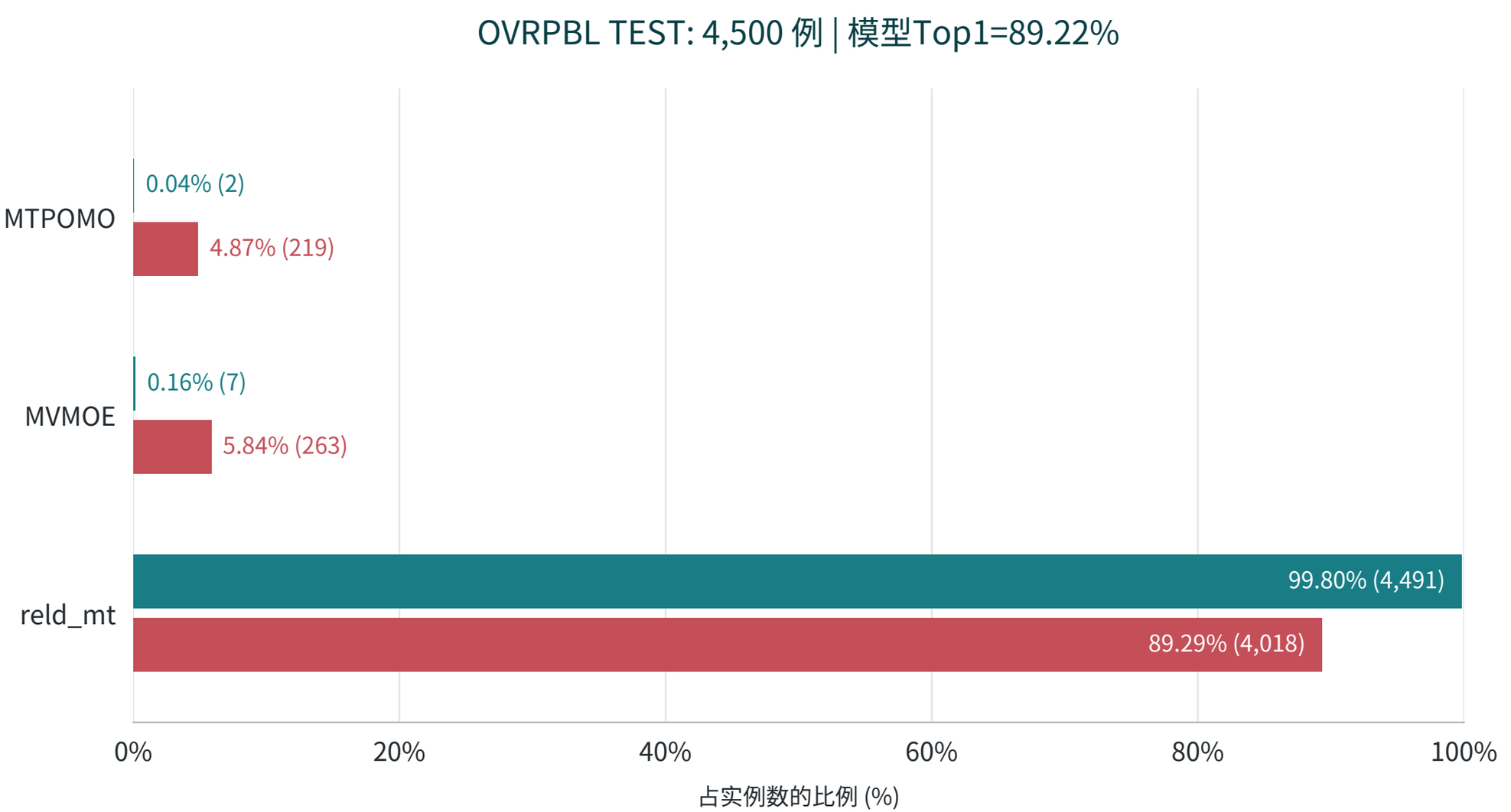
R84/B_attention_direct: stable global solver-ID score ties. Same R84 lock; LIB inference only; no training/reselection/Rxx mixing.

R84/B_attention_direct | OVRPBL | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 309b90d7a1bc5470

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

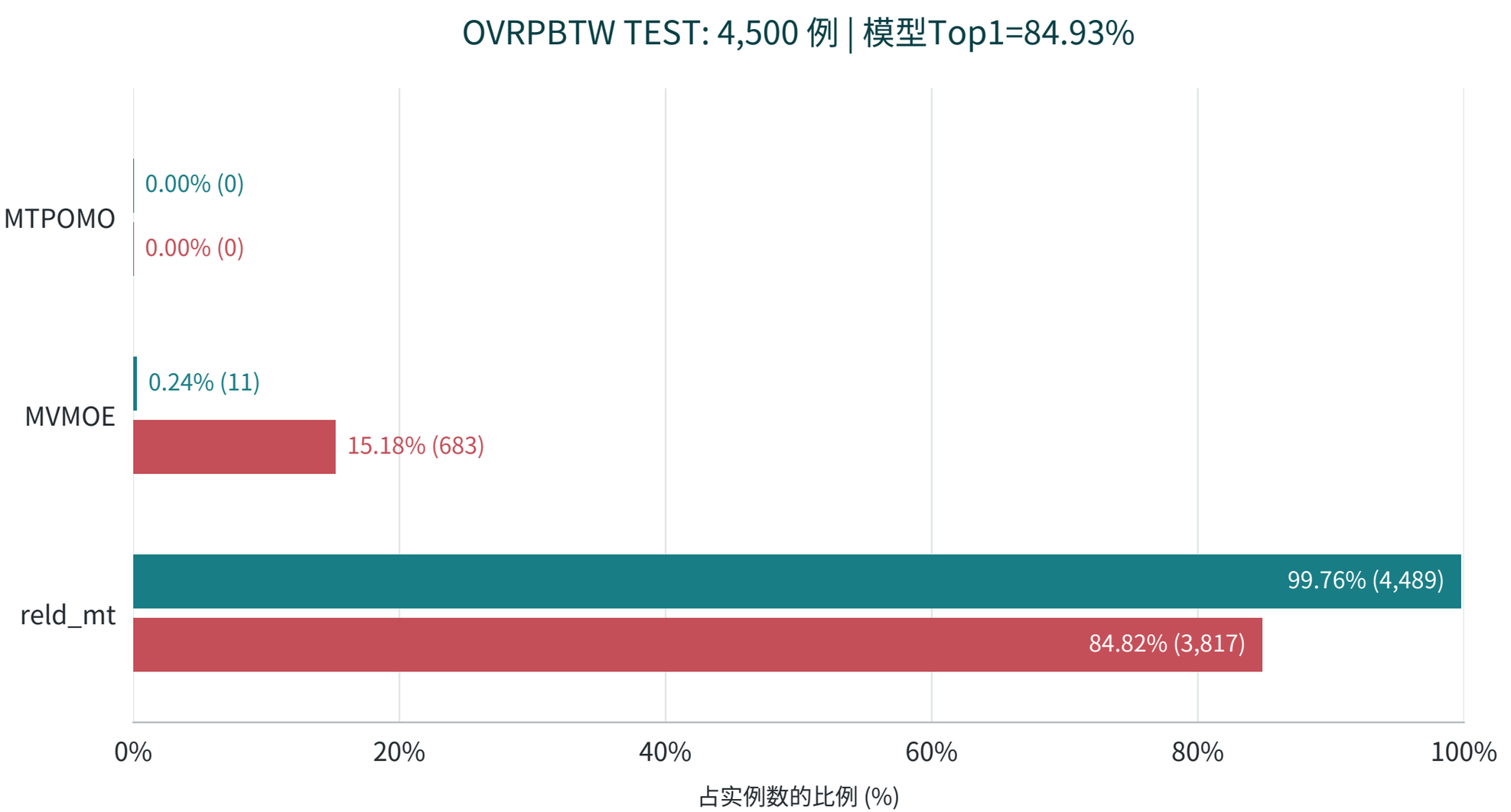
R84/B_attention_direct: stable global solver-ID score ties. Same R84 lock; LIB inference only; no training/reselection/Rxx mixing.

R84/B_attention_direct | OVRPBTW | 方法分布

同一锁定模型: epoch 26 / update 37440 / SHA256 309b90d7a1bc5470

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



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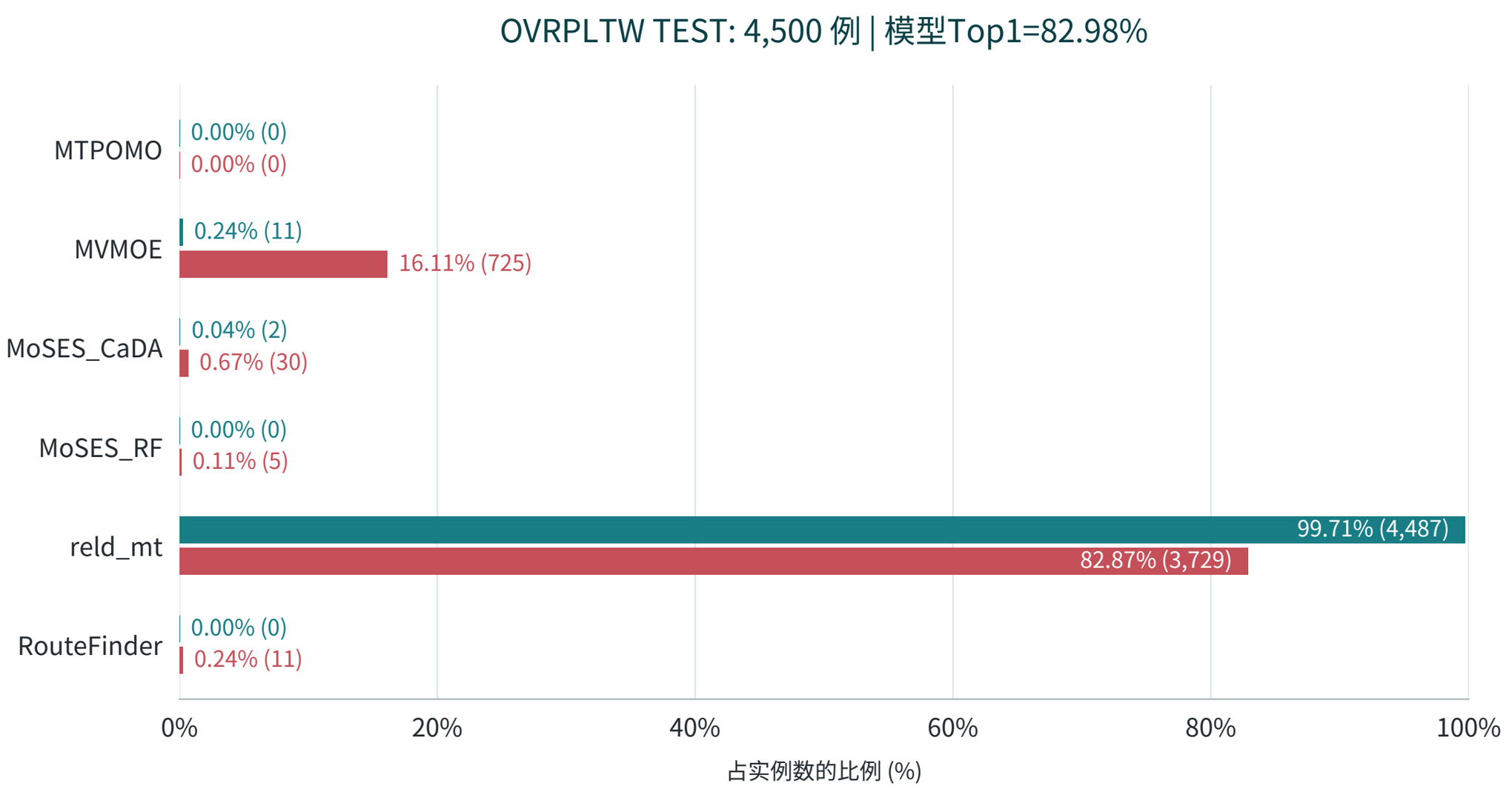
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R84/B_attention_direct | OVRPLTW | 方法分布

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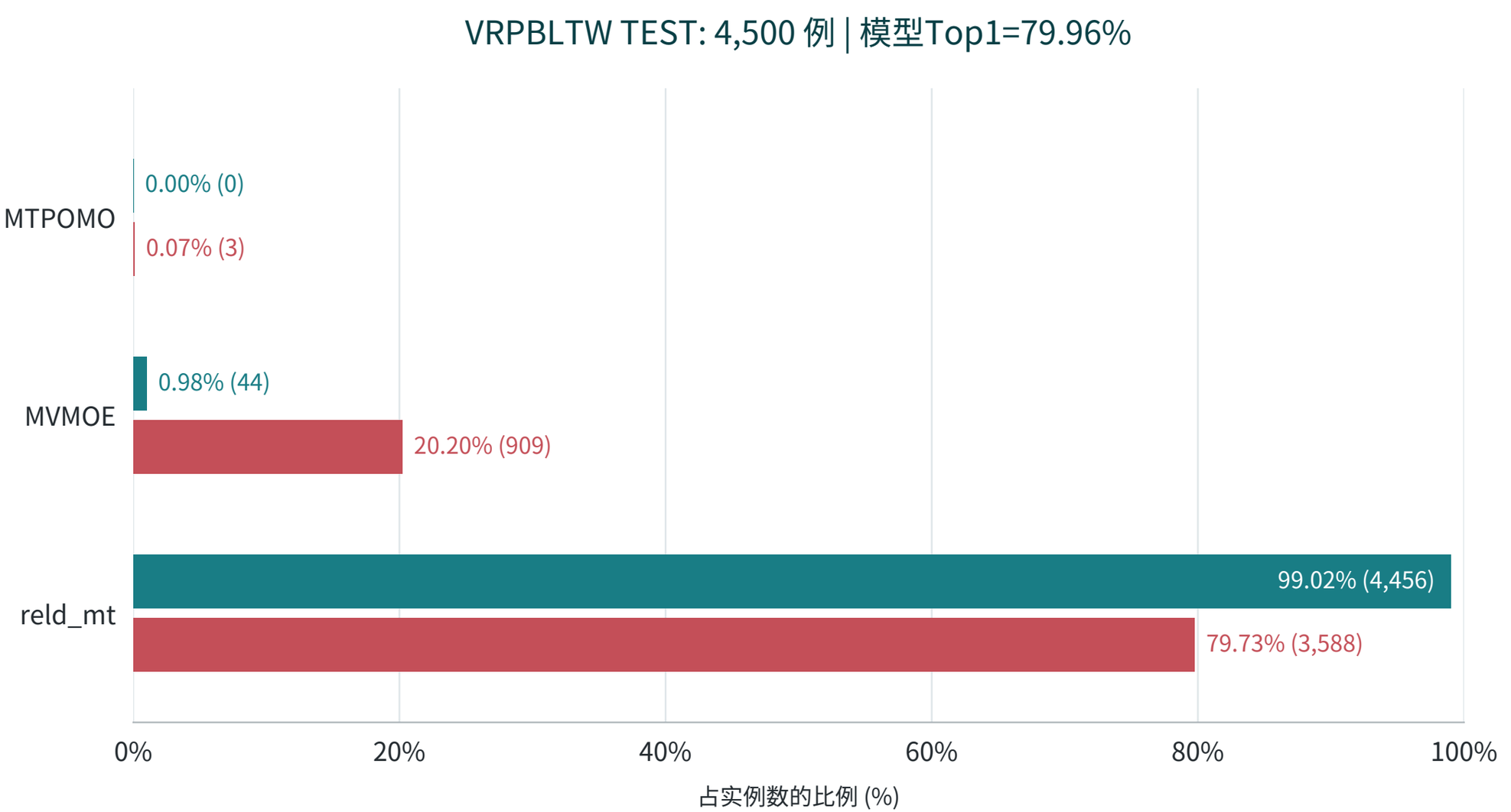
R84/B_attention_direct: stable global solver-ID score ties. Same R84 lock; LIB inference only; no training/reselection/Rxx mixing.

R84/B_attention_direct | VRPBLTW | 方法分布

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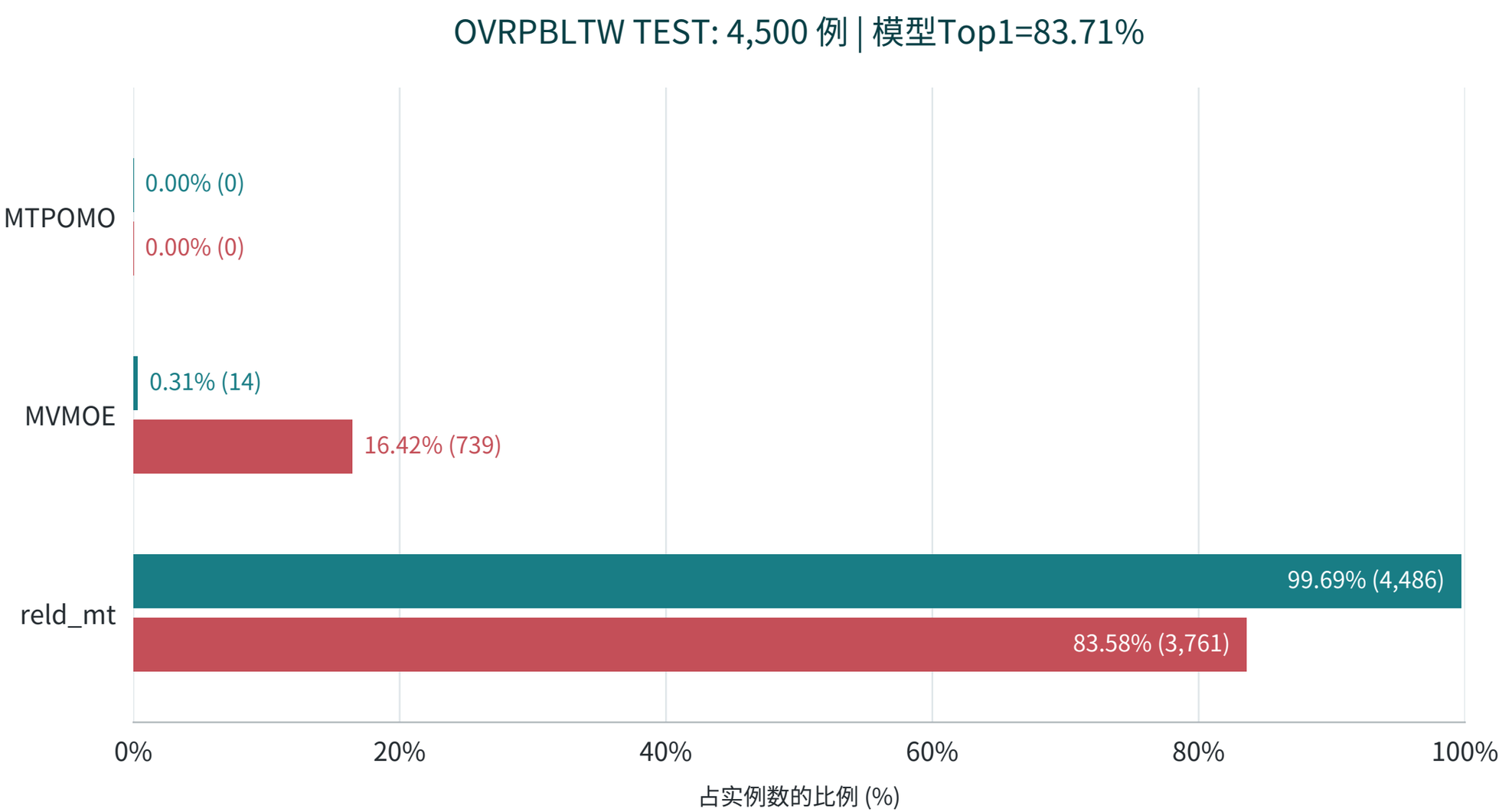
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